

Chapter 7 — Class 3

Public goods, common resources, and pollution policy

Instructor preparation

Length: 55 minutes. **Slides:** revisit 23, then 29–36. **Reading:** Chapter 7 notes, “Public Goods and the Free-Rider Problem” through the chapter review.

Learning goals: students can explain underprovision and overuse, evaluate a shared project, show how permit trading lowers cleanup cost, and distinguish the amount of cleanup from its allocation across firms.

Board setup: use three sections for the shared-project calculation, the two-firm permit table, and a cleanup graph. Write “Payments between parties are transfers” above the permit table.

0–5 minutes | Retrieve the two goods problems

Ask students to classify a widely visible light display and an open-access fishery. Require both criteria.

Write:

Good	What another user’s participation does	Likely incentive problem
Public good	Receives a shared benefit without using it up	Wait for others to pay
Common resource	Takes some of a scarce shared resource	Use more before someone else does

Say: “Both are hard to exclude people from. Rivalry explains why one tends to be underprovided and the other overused.”

5–14 minutes | Board example 1: fund a shared display? — slides 23 and 29

Assume a display can be seen by three households, nonpayers cannot be excluded, and viewing is nonrival. Values measure each household’s total willingness to pay for this one display. There are no other benefits or harms in this illustration.

Household	Value of one display
A	40

Household	Value of one display
B	35
C	25
Total	100

$$\text{Cost} = 80, \quad \text{net social benefit} = 100 - 80 = 20.$$

Ask: “Is the project worth doing?” **Answer:** yes, under these assumptions.

Ask: “Will one household pay the whole cost?” **Answer:** no household values it at 80. Joint funding can work, but voluntary funding is not assured.

Say: “A person can want the display and still prefer that other people pay. Low contributions do not necessarily mean low total benefit.”

Show a feasible cost split: A pays 30, B pays 30, C pays 20. Contributions sum to 80, and individual net gains are 10, 5, and 5. This proves mutually beneficial funding is possible; it does not prove those voluntary payments will occur.

Ask: “What if each household must pay an equal share?” Each pays about 26.67; C’s value is only 25. An efficient project need not benefit every person under every funding arrangement.

End with measurement: “The values are known in our example. In practice, people may misstate them, and we must count opportunity costs and harms such as noise.” Tax funding can solve a collection problem without automatically solving the valuation problem.

14–21 minutes | How much of the public good?

Say: “Choosing whether to provide one project is a total-benefit question. Choosing how much to provide requires marginal benefits and marginal costs.”

For one additional segment of a shared display, suppose household marginal values are 20, 15, and 10. Write:

$$\text{Community marginal benefit} = 20 + 15 + 10 = 45.$$

Everyone receives the same additional segment, so add dollar values for that unit. This is **vertical addition**. In a private-good demand calculation, by contrast, we add people’s quantities at the same price.

Now use the notes’ separate, simplified continuous model:

$$MB = 100 - 10Q, \quad MC = 40.$$

The curve is already the community total; do not add it three times. The numbers 20, 15, and 10 above only illustrate how to combine values for a single unit.

Ask students to solve:

$$100 - 10Q = 40 \Rightarrow Q^* = 6.$$

Draw: horizontal axis “Display quantity, Q ”; vertical axis “Dollars per additional unit.” Draw MB from $(0, 100)$ to $(10, 0)$ and horizontal $MC = 40$. Mark their intersection at $(6, 40)$.

Say: “Before six, more provision adds benefits greater than costs. Beyond six, more provision adds costs greater than benefits.” Treat quantity as divisible for this graph. With indivisible segments, compare each segment’s benefit and cost directly.

21–30 minutes | Board example 2: the common pasture — slides 30–34

Tell the deck’s story briefly: a pasture can support about 100 cows; 100 farmers each have one cow. Each farmer considers adding one. If all do so, there are 200 cows.

Say: “One farmer receives the gain from an extra cow. Some costs fall on everyone else through reduced grass and future productivity.”

Add this illustrative marginal calculation. These numbers are a separate teaching example, not estimates from the slides:

Extra cow’s benefit to its owner = 50

Owner’s full private marginal cost = 20

Additional damage to other users = 40.

Then calculate:

Private net gain = $50 - 20 = 30$

Social net gain = $50 - 20 - 40 = -10$.

Ask: “Why add the cow privately? Why avoid it socially?” **Answer:** the owner gains 30, but the combined effect is a loss of 10. The owner does not bear all costs.

Take two proposals to change incentives. Possible answers:

- An enforceable limit on cows or grazing days.
- A use charge reflecting marginal harm.
- Defined use rights with monitoring.
- Community rules with agreed sanctions for violations.

For each proposal ask: “Who checks whether people follow the rule?” Mention the notes’ discussion of Ostrom: communities can sometimes manage shared resources through agreed rules and enforcement. Shared use does not imply inevitable collapse.

Bridge: “Pollution also uses a shared resource: the environment’s ability to absorb emissions. A permit system limits that use.”

30–44 minutes | Board example 3: trading pollution permits — slide 35

Budget about 3 minutes for setup, 3 for equal reductions, 5 for trading and payments, and 3 for interpretation.

Assume both firms initially emit 100 units. Combined emissions are 200. The cap allows 100, so total required cleanup is 100. Firm A can clean up any of its first 100 units at 10 per unit; B can do so at 2 per unit. Damage per emitted unit is the same across firms, trading is costless, and the cap is enforced.

Write first:

$$\text{Emissions} = \text{initial emissions} - \text{cleanup}.$$

Permits authorize emissions, not cleanup. This distinction prevents most errors in the exercise.

Step 1: equal reductions

Firm	Initial emissions	Cleanup	Final emissions	Cleanup cost
A	100	50	50	500
B	100	50	50	100
Total	200	100	100	600

Ask: “Can we achieve the same total emissions with a lower resource cost?” Let students identify B as the cheaper cleaner.

Step 2: allow trading

Give each firm 50 emissions permits. Let B clean up all 100 units and sell its 50 permits to A.

Firm	Permits after trade	Cleanup	Final emissions	Cleanup cost
A	100	0	100	0
B	0	100	0	200
Total	100	100	100	200

$$\text{Resource saving} = 600 - 200 = 400.$$

Say: “Total emissions are still one hundred. The low-cost firm does more cleanup. That reduces the total cost of meeting the cap.”

Step 3: show why both firms accept a trade

Use an illustrative price of 6 per permit. A buys 50 permits for 300; B receives 300.

Firm	Cost under equal reductions	Cleanup cost after trade	Permit payment, paid (+) or received (-)	Net cost after trade	Gain from trade
A	500	0	+300	300	200
B	100	200	-300	-100	200

Explain B's negative net cost: its permit receipts exceed its cleanup expense. This is possible because B initially received valuable permits for free.

Any price strictly between 2 and 10 supports gains for both firms from this trade. The example does not determine a unique price of 6.

Ask: "Why are the 300 dollars of permit payments not an additional social cost?" **Answer:** A pays what B receives. The resource cost is the 200 spent on cleanup.

Ask: "Is a free permit costless to use?" **Answer:** no. Using it sacrifices the revenue from selling it.

Say: "Trading lowers the cost of meeting this cap. It does not tell us whether one hundred units is the best cap."

44–50 minutes | Return to the opening pollution question — slide 36

Use a separate illustrative cleanup model. Do not combine its units or numbers with the two-firm example.

Let initial pollution be 10 units and let A denote units of pollution eliminated, with $0 \leq A \leq 10$:

$$MB_{\text{cleanup}} = 12 - A, \quad MC_{\text{cleanup}} = 2 + A.$$

Solve with the class:

$$12 - A = 2 + A \Rightarrow A^* = 5.$$

Remaining pollution is $10 - 5 = 5$.

Draw: horizontal axis "Cleanup, A "; vertical axis "Dollars per cleanup unit." Draw downward MB through $(0, 12)$ and upward MC through $(0, 2)$. Mark the intersection $(5, 7)$ and full cleanup at $A = 10$.

Say: "Moving right means more cleanup and less pollution. Before five, cleanup prevents more damage than it costs. Beyond five, extra cleanup costs more than the damage it prevents."

At full cleanup, marginal benefit is 2 and marginal cost is 12 in this example. This is why zero pollution is not the efficient target here.

Ask: "Could full cleanup ever be efficient?" **Answer:** yes, if its marginal benefits are at least as large as marginal costs throughout the feasible cleanup range. The model's values determine the answer; zero is not ruled out in advance.

50–55 minutes | Exit check

Allow three minutes individually, then two minutes to review.

1. Three households value one shared warning system at 60, 50, and 40. It costs 120. Find its net social benefit and explain why voluntary funding might fail.
2. Explain in one sentence why permit trading can lower cost without raising total emissions.
3. If one more unit of cleanup prevents 3 of damage but costs 8, should it be undertaken on these numbers alone?

Answers: benefit 150, net benefit 30; households may try to benefit without contributing. Trading shifts cleanup toward lower-cost firms while the enforced cap fixes total permitted emissions. Do not undertake the extra cleanup unit: its marginal cost exceeds its marginal benefit by 5.

Closing sentence: “For each policy, ask which cost or benefit is missing from private decisions, and whether the proposed rule changes that incentive.”

Pacing and optional depth

If behind: omit the equal-share display discussion and the public-good graph, but keep the vertical-addition explanation. In permit trading, calculate costs aloud and omit the final net-cost table if needed. Protect the final cleanup comparison and exit check.

Optional independent practice: two firms each emit 80 units; the cap is 80; constant cleanup costs are 9 for A and 3 for B. Equal reductions cost $40(9) + 40(3) = 480$. Having B clean up all 80 costs 240. With 40 permits initially assigned to each, B sells 40 to A. At an illustrative price of 6, A pays 240 instead of spending 360 on cleanup; B spends 240 and receives 240 instead of spending 120. Each gains 120. Total emissions remain 80.

Optional discussion: with rising marginal cleanup costs, lowest-cost allocation generally equalizes firms’ marginal cleanup costs for interior choices. With constant costs in today’s example, concentrate cleanup at B until its capacity is exhausted. Location-dependent pollution harm can make emissions at the two plants environmentally different even if the total is unchanged.

Misconceptions to listen for: public goods have no cost; a valuable project automatically receives voluntary funding; common resources are nonrival; a permit authorizes cleanup; trading changes the cap; zero pollution is always or never efficient.